

Police Management System

Use Cases Document

1. All Users

UC1: Login / Logout

Summary	Any authorized user logs into the system using their credentials and logs out when finished.
Dependency	None
Actors	Primary: All Users (Police Officer, Detective, Supervisor, Administrator)
Preconditions	1. User must have valid credentials issued by the Administrator. 2. System must be accessible via web browser.
Main Sequence	1. User navigates to the login page. 2. System displays the login form (Username, Password). 3. User enters their username and password. 4. User clicks the Login button. 5. System validates the credentials. 6. System grants access and redirects the user to their role-based dashboard. 7. User performs their tasks. 8. User clicks Logout. 9. System ends the session and returns to the login page.
Alternative Sequence	1. Invalid credentials: System displays an error message. User may retry. 2. Account locked/inactive: System displays an error indicating the account is not active and prompts contacting the Administrator.
Non-Functional Requirements	1. Login should complete within 3 seconds under normal load. 2. Passwords must be encrypted (hashed) in the database. 3. System must enforce session timeout after inactivity.
Postconditions	1. User is authenticated and their session is active. 2. On logout, the session is terminated and no sensitive data remains accessible.

UC2: View Dashboard

Summary	After logging in, any user can view a dashboard with statistics relevant to their role.
Dependency	UC1 (User logs in)
Actors	Primary: All Users (Police Officer, Detective, Supervisor, Administrator)
Preconditions	1. User is authenticated and logged in.

	2. User has a role-specific dashboard configured.
Main Sequence	1. User is redirected to the dashboard after successful login. 2. System retrieves relevant statistics from the database. 3. System displays the dashboard with data such as: Open Cases, Cases Under Investigation, Recent Incidents, Assigned Officers, Departments (for Admin).
Alternative Sequence	1. No data available: System displays a message stating that no statistics are currently available. 2. Data loading error: System shows an error message and provides an option to refresh.
Non-Functional Requirements	1. Dashboard must load within 5 seconds. 2. Statistics must reflect real-time data. 3. Dashboard layout must be responsive and accessible from any device.
Postconditions	1. The dashboard is displayed with up-to-date statistics for the user's role. 2. User can navigate to specific sections from the dashboard.

2. Police Officer

UC3: Create Incident Report

Summary	A Police Officer creates a new incident report to document a reported crime in the system.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	1. Police Officer is logged in and has permission to create incident reports. 2. The system is operational.
Main Sequence	1. Officer navigates to the 'Reports' section. 2. Officer clicks 'Create New Incident Report'. 3. System displays the incident report form. 4. Officer enters the required details: location, crime type, description, date and time. 5. Officer submits the form. 6. System validates all required fields. 7. System saves the incident report in the database. 8. System automatically generates a new case linked to the incident report. 9. System displays a confirmation message with the Case ID.
Alternative Sequence	1. Missing required fields: System highlights the empty fields and displays an error message. Officer must complete them before submitting.

	2. System error during save: System displays an error and prompts the officer to try again.
Non-Functional Requirements	1. The form must be simple and complete within 5 minutes. 2. System must validate data formats (e.g., date/time format, text length). 3. Data must be stored securely with access restricted by role.
Postconditions	1. A new incident report is saved in the system. 2. A new case is automatically created and linked to the report. 3. The case appears in the Cases list with 'Open' status.

UC4: Update Case Status

Summary	A Police Officer updates the status of an existing case (Open, Under Investigation, or Closed).
Dependency	UC1 (User logs in), UC3 (Create Incident Report)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. The case exists in the system and is assigned to or accessible by the officer.
Main Sequence	1. Officer navigates to the 'Cases' section. 2. Officer selects the case to update. 3. System displays the case details. 4. Officer selects a new status from the dropdown (Open, Under Investigation, Closed). 5. Officer saves the update. 6. System validates the status change. 7. System saves the updated status in the database. 8. System confirms the update with a success message.
Alternative Sequence	1. Unauthorized access: If the officer tries to update a case not assigned to them, the system denies the action. 2. Same status selected: System notifies the user that the status is unchanged.
Non-Functional Requirements	1. Status update should be saved within 2 seconds. 2. Status change history should be logged for audit purposes. 3. Access control must prevent unauthorized status changes.
Postconditions	1. The case status is updated in the database. 2. The change is reflected immediately in the case list and dashboard.

UC5: Upload Evidence

Summary	A Police Officer or Detective uploads evidence files related to an existing case.
Dependency	UC1 (User logs in), UC3 (Create Incident Report)

Actors	Primary: Police Officer Secondary: Detective
Preconditions	1. User is logged in with permission to upload evidence. 2. A case already exists in the system.
Main Sequence	1. User navigates to the specific case. 2. User clicks 'Upload Evidence'. 3. System displays the evidence upload form. 4. User selects the evidence file(s) and enters a description. 5. User submits the upload. 6. System validates the file format and size. 7. System stores the evidence and links it to the case. 8. System confirms the upload with a success message.
Alternative Sequence	1. Invalid file format or size exceeded: System displays an error and requests a valid file. 2. Upload failure: System shows an error and allows retry.
Non-Functional Requirements	1. Supported file formats must be defined (e.g., PDF, JPG, MP4). 2. Maximum file size per upload should be enforced. 3. Evidence files must be stored securely with restricted access.
Postconditions	1. Evidence is stored securely in the system. 2. Evidence is linked to the correct case and visible in case details.

UC6: Record Arrest Information

Summary	A Police Officer records the details of an arrest and links them to the relevant case.
Dependency	UC1 (User logs in), UC3 (Create Incident Report)
Actors	Primary: Police Officer
Preconditions	1. Officer is logged in. 2. A case exists in the system. 3. An arrest has been made related to an existing case.
Main Sequence	1. Officer navigates to the relevant case. 2. Officer selects 'Add Arrest Record'. 3. System displays the arrest information form. 4. Officer fills in: arrested person's name, date of arrest, location, charges. 5. Officer submits the form. 6. System validates the input. 7. System saves the arrest record and links it to the case. 8. System confirms with a success message.
Alternative Sequence	1. Incomplete information: System highlights missing required fields. 2. Case not found: System displays an error and prevents submission.
Non-Functional Requirements	1. Arrest records must be immutable once submitted (controlled edit process). 2. Data must comply with law enforcement data regulations.

	3. Records must be audit-logged.
Postconditions	1. Arrest record is saved and linked to the case. 2. Case details are updated to reflect the arrest.

3. Detective

UC7: Assign Suspect to Case

Summary	A Detective links a suspect to an existing case by entering suspect details into the system.
Dependency	UC1 (User logs in), UC3 (Create Incident Report)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. A case exists in the system. 3. There is a potential suspect related to the case.
Main Sequence	1. Detective navigates to the relevant case. 2. Detective selects 'Add Suspect'. 3. System displays the suspect information form. 4. Detective enters: name, age, physical description, and associated evidence. 5. Detective submits the form. 6. System validates the input. 7. System saves the suspect information and links it to the case. 8. System confirms with a success message.
Alternative Sequence	1. Case does not exist: System displays an error and prevents submission. 2. Missing required fields: System highlights empty fields and requires completion.
Non-Functional Requirements	1. Suspect data must be stored securely with restricted access. 2. Suspect records must be linked clearly to the correct case. 3. Input validation must prevent entry of invalid data formats.
Postconditions	1. Suspect information is saved and linked to the case. 2. Suspect appears in the case details under the Suspects section.

UC8: View and Analyze Case Details

Summary	A Detective reviews all information associated with a case — incident reports, suspects, and evidence — to determine the investigative direction.
Dependency	UC1 (User logs in)
Actors	Primary: Detective
Preconditions	1. Detective is logged in. 2. One or more cases are assigned to the detective.
Main Sequence	1. Detective navigates to the Cases section. 2. Detective selects a case assigned to them. 3. System retrieves and displays all case details: crime type, status, assigned officer, suspects, evidence, and incident reports. 4. Detective reviews the information. 5. Detective may add notes or initiate follow-up actions (e.g., assign suspect, upload evidence).
Alternative Sequence	1. Case has no data yet: System displays the case with empty sections and a note indicating no data has been added. 2. Unauthorized access to a case not assigned to the detective: System denies access.
Non-Functional Requirements	1. Case details must load within 3 seconds. 2. All sensitive case data must be accessible only to authorized roles. 3. The view must display all sections clearly: Suspects, Evidence, Incident Reports.
Postconditions	1. Detective has full visibility into all case-related data. 2. Detective can proceed to take further investigative actions from the case view.

4. Supervisor

UC9: Assign Officer to Case

Summary	A Supervisor assigns a Police Officer to a specific case to ensure proper investigative coverage.
Dependency	UC1 (User logs in), UC3 (Create Incident Report)
Actors	Primary: Supervisor
Preconditions	1. Supervisor is logged in. 2. An active case exists in the system. 3. One or more Police Officers are registered in the system.
Main Sequence	1. Supervisor navigates to the active case. 2. Supervisor selects 'Assign Officer'. 3. System displays a list of available Police Officers. 4. Supervisor selects the appropriate officer. 5. Supervisor confirms the assignment. 6. System validates the selection.

	- System saves the assignment and notifies the assigned officer. - System confirms with a success message.
Alternative Sequence	- No available officers: System displays a message indicating all officers are currently assigned. - Case already has an assigned officer: System asks the Supervisor to confirm the reassignment.
Non-Functional Requirements	- Assignment must be saved and reflected within 2 seconds. - The assigned officer must receive a notification. - Officer assignment history must be logged.
Postconditions	- The selected officer is assigned to the case. - The case details are updated with the assigned officer's information. - The officer can now access and manage the case.

5. Administrator

UC10: Manage Users and Roles

Summary	The Administrator creates, updates, deactivates, and manages user accounts and their role assignments within the system.
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - The system is operational.
Main Sequence	- Administrator navigates to the 'Officers' (Users) management section. - To add a user: Administrator clicks 'Add', fills in Username, Password, Department, Name, Surname, Birthdate, and Role. Clicks Save. System validates and creates the account. - To edit a user: Administrator selects a user, clicks 'Edit', modifies the required fields, and saves. System updates the record. - To delete/deactivate a user: Administrator selects a user and clicks 'Delete'. System prompts for confirmation. Administrator confirms. System deactivates or removes the account. - System confirms each action with a success message.
Alternative Sequence	- Duplicate username: System prevents creation and displays an error. - Missing required fields: System highlights missing fields. - Attempt to delete active user with ongoing cases: System warns the administrator and requests confirmation.

Non-Functional Requirements	1. User management operations must complete within 3 seconds. 2. Password must be hashed before storage. 3. All user management actions must be audit-logged.
Postconditions	1. New user account is created with the correct role and department. 2. Updated user information is reflected in the system. 3. Deactivated user can no longer log in.

UC11: Manage Departments

Summary	The Administrator adds, edits, and manages police department records to organize officers by their respective units.
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in. 2. System is operational.
Main Sequence	1. Administrator navigates to the 'Departments' section. 2. To add: Administrator clicks 'Add', enters the Department Name, and clicks Save. System saves the new department. 3. To edit: Administrator selects a department, clicks 'Edit', modifies the name, and saves. 4. To delete: Administrator selects a department and clicks 'Delete'. System checks if officers are assigned. If no officers assigned, department is deleted. If officers are assigned, system warns the administrator. 5. System confirms each action with a success message.
Alternative Sequence	1. Department name already exists: System displays a duplicate error. 2. Attempt to delete department with assigned officers: System warns and prevents deletion until officers are reassigned.
Non-Functional Requirements	1. Department operations should complete within 2 seconds. 2. Department names must be unique within the system. 3. Changes must be immediately visible to all relevant users.
Postconditions	1. New department is saved and available for officer assignment. 2. Updated department information is reflected in the system. 3. Deleted department is removed from all dropdown lists.

UC12: Monitor System Statistics

Summary	The Administrator monitors real-time analytics and statistics across the system including cases, users, departments, and incidents.
Dependency	UC1 (Administrator logs in), UC2 (View Dashboard)
Actors	Primary: Administrator
Preconditions	1. Administrator is logged in.

	2. Data has been entered into the system by users.
Main Sequence	1. Administrator navigates to the Dashboard. 2. System retrieves all system-wide statistics. 3. System displays: number of active departments, total assigned officers, open cases, cases under investigation, recent incidents. 4. Administrator reviews the data and monitors performance. 5. Administrator can navigate to specific sections for more detailed information.
Alternative Sequence	1. No data available: System displays zero values and a note indicating the system has no recorded data. 2. Data loading failure: System shows an error and provides a refresh option.
Non-Functional Requirements	1. Dashboard statistics must reflect real-time data. 2. Statistics page must load within 5 seconds. 3. Data displayed must be accurate and sourced from the live database.
Postconditions	1. Administrator has a comprehensive, real-time overview of all system operations. 2. Administrator can take corrective action based on the statistics displayed.